

## Row-vector convention

$$\mathbf{r} = \mathbf{f} \mathbf{H}$$

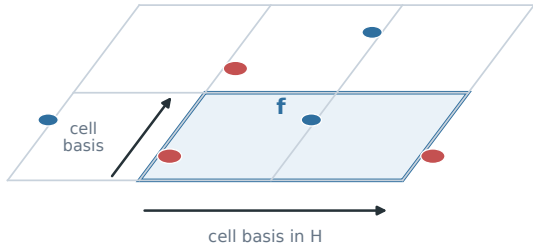
fractional  $\mathbf{f}$ : dimensionless

Cartesian  $\mathbf{r}$ : angstrom

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$$\mathbf{f} \sim \mathbf{f} + \mathbf{n}$$

integer lattice  $\mathbf{n}$ ; wrap to  $[0, 1)$



atomic species + fractional positions + cell matrix define the periodic structure